

04

Determinant

Transforms a square matrix into a scalar; area, volume ...

4.1 The 2×2 Matrix Determinant



Core skills you'll master in this section:

- ▶ Characteristics of the determinant operation: exclusive to square matrices, and the result is a scalar.
- ▶ The determinant of a 2×2 matrix represents the signed area of a parallelogram.
- ▶ Identity matrix: the unit square.
- ▶ Diagonal square matrix with identical diagonal elements: the column vectors form a square.
- ▶ Diagonal square matrix with different diagonal elements: the column vectors form a rectangle.
- ▶ Upper triangular matrix: the column vectors form a parallelogram sheared along the horizontal axis.
- ▶ Lower triangular matrix: the column vectors form a parallelogram sheared along the vertical axis.
- ▶ A square matrix with a determinant of 0 is not invertible (this will be covered in detail in the next chapter).

Let's put the most important point first: the determinant of a square matrix is a scalar value. This statement has two key points: first, only square matrices have determinants; second, a determinant is a scalar value.

Determinants have wide-ranging uses in mathematics and applied fields.

First, the determinant can be used to determine whether a matrix is invertible. A square matrix is invertible only when its determinant is nonzero; when the determinant is zero, the square matrix is called a singular matrix, and it is not invertible. This property is especially important when solving systems of linear equations, computing the adjugate matrix, and computing the matrix inverse. Later in this book, we will devote a dedicated section to the inverse matrix.

Second, in geometry, the determinant is closely tied to changes in area and volume. The determinant value of a linear transformation represents the scaling factor that transformation applies to area or volume. For example, in the two-dimensional plane, the determinant can measure how a linear transformation changes the area of a parallelogram; in three-dimensional space, it determines the change in volume of a parallelepiped. This geometric perspective is an important tool for understanding determinants throughout this chapter!

In addition, applications of determinants in linear algebra also include determining the linear dependence of a set of vectors and computing the eigenvalues of a matrix. These concepts are essential for a deep understanding of linear transformations, differential equations, computer graphics, machine learning, and other fields.

However, computing and understanding determinants is not easy. To help you grasp the essence of the determinant more intuitively, we'll start from a geometric perspective. This section uses a parallelogram to understand the determinant of a 2×2 matrix, and the next section extends this to the determinant of a 3×3 matrix using a parallelepiped.

The Determinant of a 2×2 Matrix

Given the following 2×2 matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Matrix $\mathbf{A}$'s determinant is

$$\det(\mathbf{A}) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

First, throughout this book we'll commonly use $\det()$ as the determinant operator; $\det$ is taken from the first three letters of the word determinant.

When discussing determinants, keep in mind that they apply only to square matrices; for a non-square matrix, the determinant is not defined. As a reminder, a square matrix is one whose number of rows equals its number of columns — that is, an $n \times n$ matrix.

Simply put, an important characteristic of the determinant operation is that it maps a square matrix to a scalar value.

This means that although a square matrix itself contains multiple elements, the result of computing its determinant is always a single numerical value. This value carries rich geometric and algebraic meaning in mathematics and its applications.

For a 2×2 matrix, we'll soon see that its determinant value is closely related to the area of a particular parallelogram in the plane. This geometric perspective lets us more intuitively understand how the determinant is computed and the role it plays in spatial transformations.

The Area of a Parallelogram

Let's start with the 2×2 square matrix $\mathbf{A}$, gradually construct a parallelogram, compute its area, and then understand the geometric meaning of the determinant.

First, split the 2×2 square matrix $\mathbf{A}$ into two column vectors

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} a \\ c \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$$

Let's plot the column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ in the plane, with $\mathbf{a}_1$ and $\mathbf{a}_2$'s starting points at the origin. This parallelogram is exactly the parallelogram rule we mentioned when learning vector addition.

As shown in Figure 1, the four vertices of the parallelogram we constructed are located, respectively, at:

- 1) The origin $(0, 0)$, i.e., the starting point of vectors $\mathbf{a}_1$ and $\mathbf{a}_2$;
- 2) The endpoint of the first column vector $\mathbf{a}_1$: (a, c) ;
- 3) The endpoint of the second column vector $\mathbf{a}_2$: (b, d) ;
- 4) The endpoint of the sum of the two vectors: $(a + b, c + d)$.

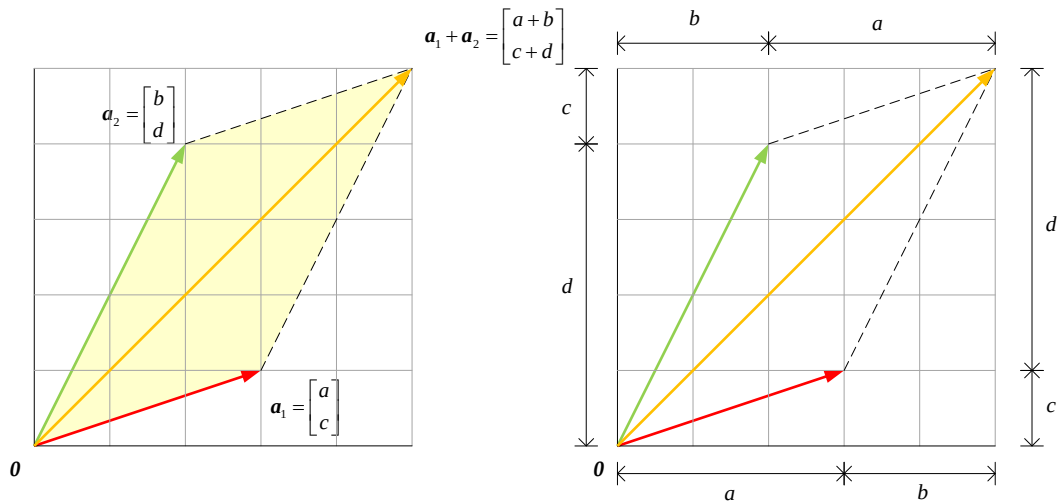


Figure 1. Constructing the Parallelogram

To compute the area of this parallelogram, let's first construct the rectangle shown in Figure 2, and then obtain the final area of the parallelogram by subtracting the extra parts.

First, construct a parallelogram with vertices at $(0, 0)$, (a, c) , (b, d) , and $(a + b, c + d)$. We can see that this parallelogram is contained within a circumscribing rectangle whose diagonal vertices are $(0, 0)$ and $(a + b, c + d)$,

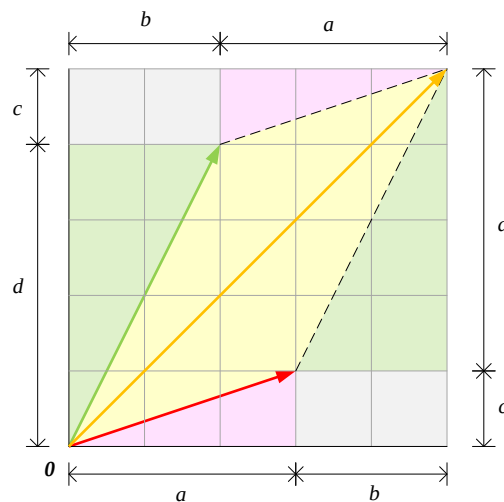


Figure 2. Constructing and Cutting the Rectangle

However, this rectangle contains parts that don't belong to the parallelogram, including four right triangles and two small rectangles:

Four triangles: located respectively at the lower-left and upper-right corners of the rectangle, corresponding to the pink and green regions in Figure 2.

Two small rectangles: located respectively at the upper-left and lower-right corners of the rectangle, corresponding to the gray regions in Figure 2.

As shown in Figure 3, the area of the parallelogram can be computed by subtracting these parts from the area of the circumscribing rectangle.

In the end, the computed result turns out to be exactly equal to the 2×2 square matrix $\mathbf{A}$'s determinant

$$\det(\mathbf{A}) = ad - bc$$

This is precisely the intuitive geometric meaning of the 2×2 determinant.

⚠ Note that in this section, area can be positive or negative; in the next section, volume can be positive or negative.

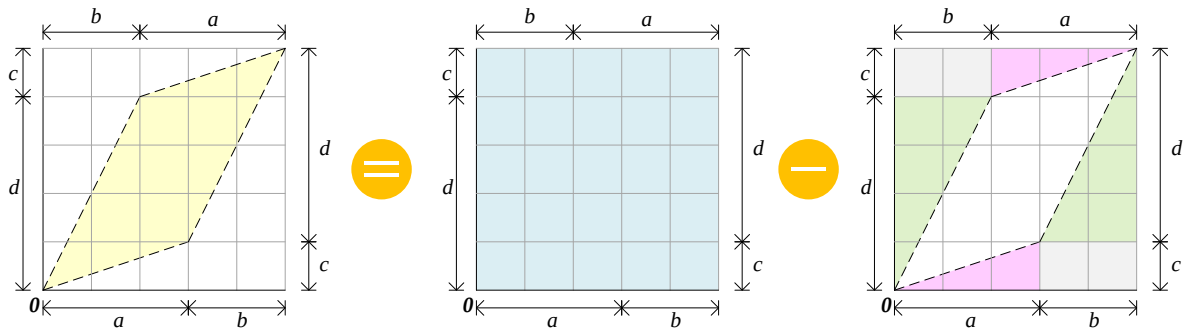


Figure 3. Computing the Area of the Parallelogram

Next, let's look at a few special 2×2 square matrices $\mathbf{A}$, and compute their determinants.

The Identity Matrix

If the 2×2 square matrix $\mathbf{A}$ is the identity matrix, write $\mathbf{A}$ as two column vectors

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The determinant is

$$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \times 1 - 0 \times 0 = 1$$

This tells us that the determinant of the identity matrix is 1.

As shown in Figure 4 (a), $\mathbf{a}_1$ and $\mathbf{a}_2$ form the unit square, whose area is exactly 1.

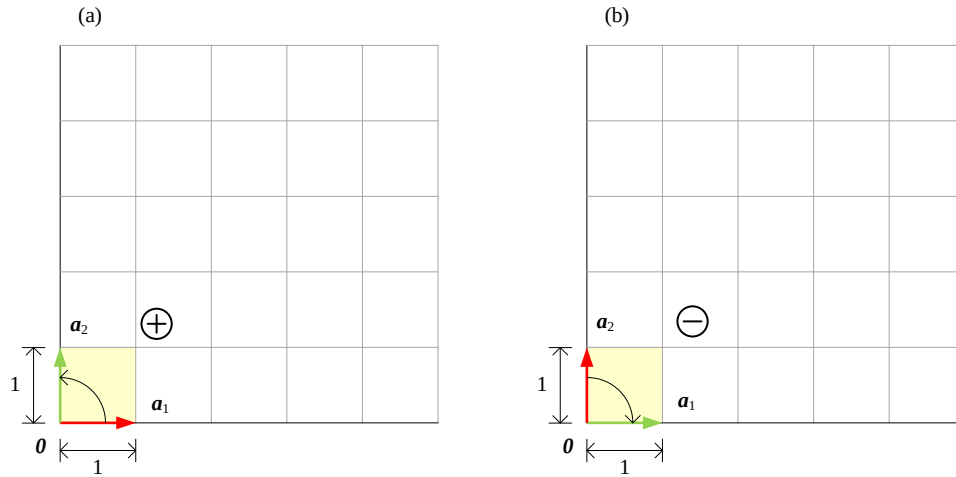


Figure 4. The Unit Square, First Set

As shown in Figure 4 (a), looking counterclockwise (within 180 degrees), the red vector $\mathbf{a}_1$ comes first, and the green vector $\mathbf{a}_2$ comes second.

By default in this section, red represents the first column vector $\mathbf{a}_1$, and green represents the second column vector $\mathbf{a}_2$.

If we swap the identity matrix's two columns, we get

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This square matrix's determinant is

$$\det \begin{pmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{pmatrix} = 0 \times 0 - 1 \times 1 = -1$$

As shown in Figure 4 (b), the parallelogram formed by these two vectors is still the unit square, so the absolute value of the area is still 1.

However, after swapping the column vectors, the originally counterclockwise vector order becomes clockwise, which is equivalent to flipping the coordinate system, causing the determinant's value to become -1.

Swapping two column vectors is equivalent to flipping the coordinate system once, so the determinant's value changes sign.

Let's look at two more examples.

As shown in Figure 5 (a), this 2×2 square matrix also forms a unit square, so the area is 1; however, viewed counterclockwise, the green vector leads the red vector, so the determinant is negative, that is

$$\det \begin{pmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \end{pmatrix} = -1 \times 1 - 0 \times 0 = -1$$

Figure 5 (b)'s corresponding 2×2 square matrix has determinant

$$\det \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = (-1) \times (-1) - 0 \times 0 = 1$$

? Please analyze on your own, from a geometric perspective, Figure 5 (b)'s corresponding 2×2 square matrix's magnitude and sign.

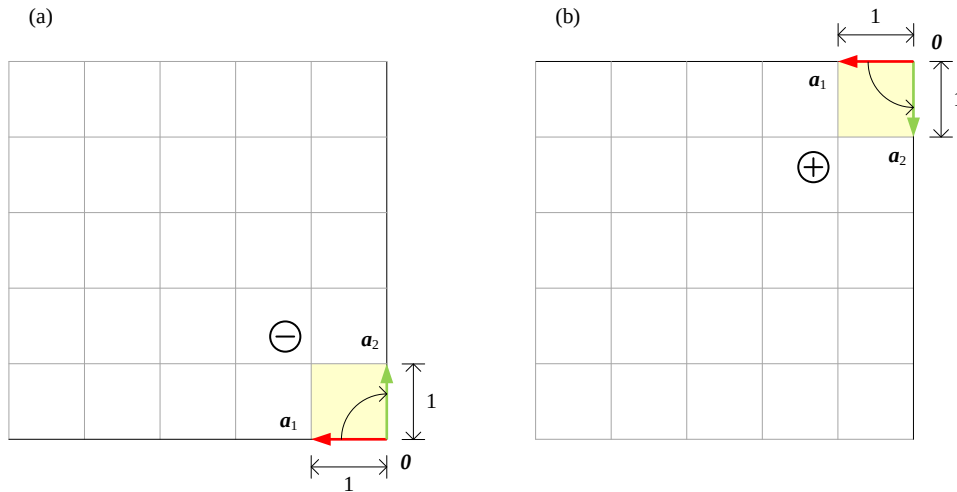


Figure 5. The Unit Square, Second Set

Diagonal Square Matrix with Identical Main-Diagonal Elements

The 2×2 square matrix $\mathbf{A}$ is a diagonal square matrix, with identical elements on the main diagonal, for example

$$\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

The determinant of the above matrix is

$$\det \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = 3 \times 3 - 0 \times 0 = 9$$

As shown in Figure 6 (a), this 2×2 square matrix $\mathbf{A}$'s corresponding geometric shape is a square.

? Please analyze on your own Figure 6 (b)'s corresponding 2×2 square matrix $\mathbf{A}$'s determinant.

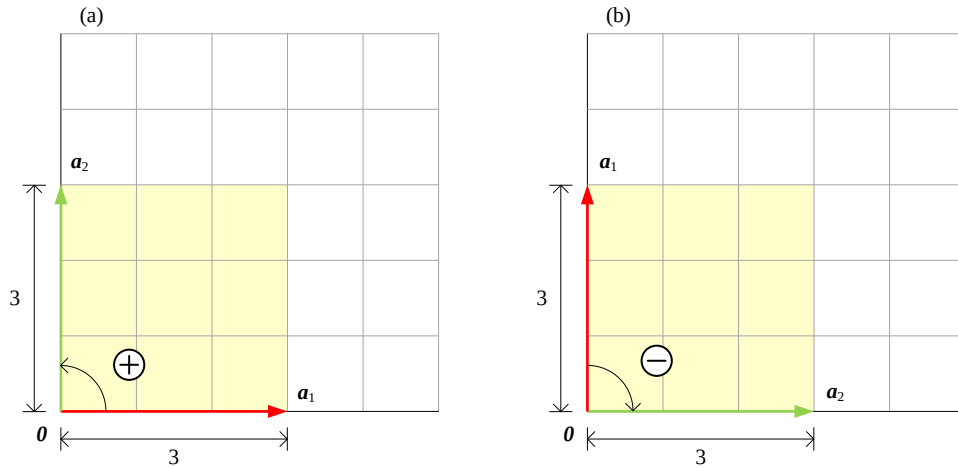


Figure 6. The Square

Diagonal Square Matrix with Different Main-Diagonal Elements

When the 2×2 square matrix A is a diagonal square matrix but its main-diagonal elements differ, A 's corresponding geometric shape is a rectangle.

As shown in Figure 7 (a), this 2×2 square matrix A 's area is "length $\times$ width," that is

$$\det \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} = 8$$

The effect of a diagonal square matrix is like scaling a shape in the horizontal and vertical directions.

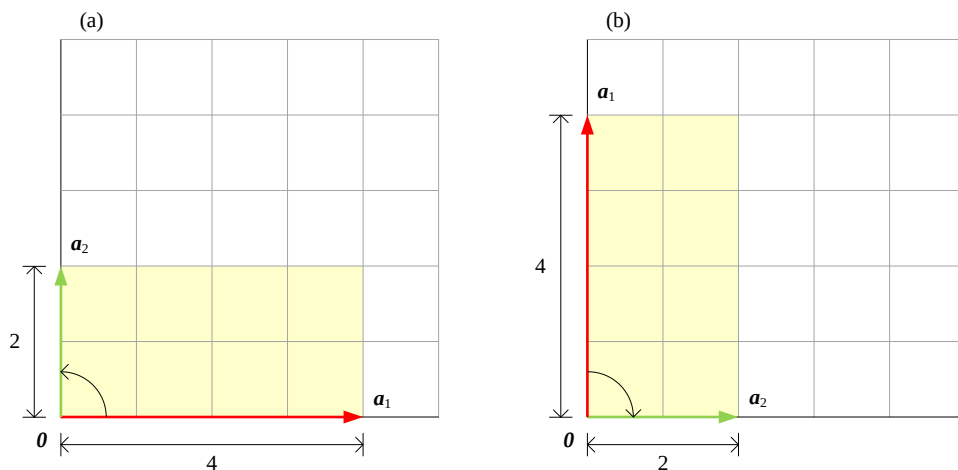


Figure 7. The Rectangle

From the above examples, we can see that the determinant of a diagonal square matrix equals the product of its main-diagonal elements, that is

$$\det \begin{pmatrix} d_1 & & \\ & d_2 & \\ & & \ddots \\ & & & d_n \end{pmatrix} = d_1 \times d_2 \times \cdots \times d_n$$

This also tells us that if any main-diagonal element of a diagonal square matrix is 0, the determinant of that diagonal square matrix is 0.

Upper Triangular Matrix

Let's now turn our attention to the upper triangular matrix.

As shown in Figure 8 (a), this 2×2 upper triangular matrix's first column vector $\mathbf{a}_1$'s direction is always parallel to the x_1 axis; its x_1 axis component is equivalent to the parallelogram's base length.

As long as the second column vector $\mathbf{a}_2$'s x_2 axis component is fixed — corresponding to the parallelogram's height — no matter how $\mathbf{a}_2$'s x_1 axis component changes, the parallelogram's area always stays the same.

In other words, $\mathbf{a}_2$'s x_1 axis component only affects how slanted the parallelogram is, not its area.

Figure 8 (a)'s corresponding 2×2 upper triangular matrix has determinant

$$\det \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} = 6$$

Figure 8 (b)'s corresponding 2×2 upper triangular matrix's determinant is clearly also 6

$$\det \begin{pmatrix} 3 & 2 \\ 0 & 2 \end{pmatrix} = 6$$

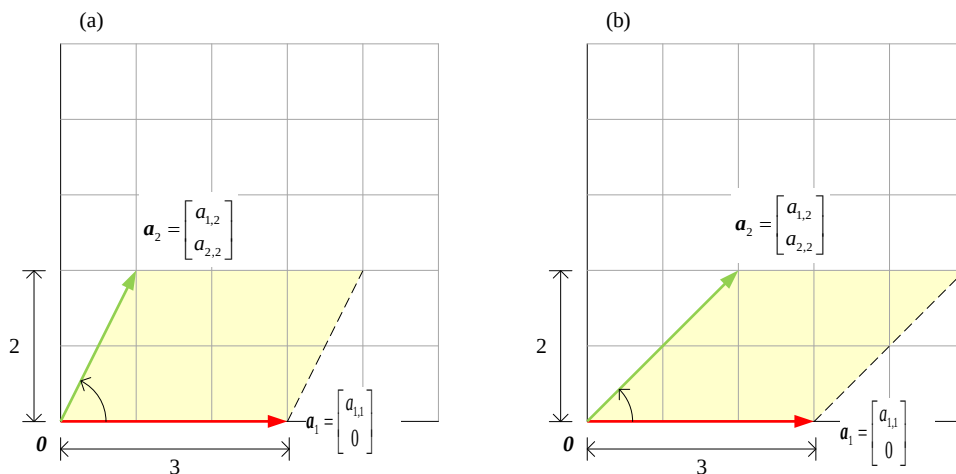


Figure 8. Upper Triangular Matrix

Geometrically speaking, Figure 8's transformation is equivalent to applying a horizontal shear (plus overall scaling) to the parallelogram — that is, without changing the height, stretching or compressing along the x_1 axis direction, tilting the parallelogram but leaving its area unchanged.

Therefore, the determinant of a 2×2 upper triangular matrix is determined solely by its main-diagonal elements, since they correspond respectively to the parallelogram's base length and vertical height, directly determining its final area.

The above examples tell us that the determinant of an upper triangular matrix equals the product of its main-diagonal elements, that is

$$\det \begin{bmatrix} d_1 & * & * & * \\ & d_2 & * & * \\ & & \ddots & \vdots \\ & & & d_n \end{bmatrix} = d_1 \times d_2 \times \cdots \times d_n$$

Similar to a diagonal square matrix, if any main-diagonal element of an upper triangular matrix is 0, the determinant of that upper triangular matrix is 0.

Lower Triangular Matrix

Having covered the determinant of the upper triangular matrix, understanding the determinant of the lower triangular matrix is now easy.

As shown in Figure 9 (a), in this 2×2 lower triangular matrix, the second column vector $\mathbf{a}_2$'s direction is always parallel to the x_2 axis.

And the first column vector $\mathbf{a}_1$'s x_1 component determines the parallelogram's height. The parallelogram's area has nothing to do with $\mathbf{a}_1$'s x_2 component!

Figure 9 (a)'s corresponding lower triangular matrix,

$$\det \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = 6$$

Figure 9 (b) corresponds to

$$\det \begin{bmatrix} 2 & 0 \\ 2 & 3 \end{bmatrix} = 6$$

Geometrically speaking, this transformation is equivalent to applying a vertical shear (plus scaling) to the parallelogram.

The determinant of a 2×2 lower triangular matrix is likewise determined solely by its main-diagonal elements, since they correspond respectively to the parallelogram's base length and vertical height, directly determining its final area.

Similar to the diagonal square matrix and upper triangular matrix, the determinant of a lower triangular matrix equals the product of its main-diagonal elements.

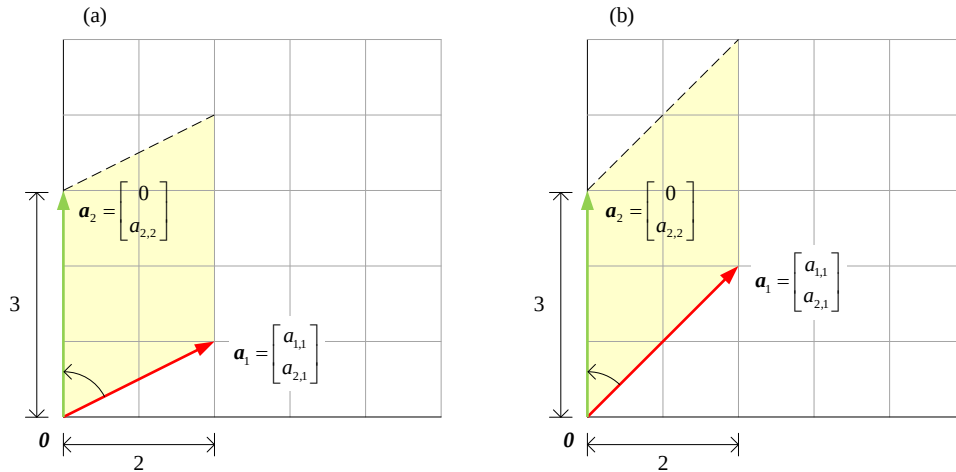


Figure 9. Lower Triangular Matrix

Code 1 visualizes this section's matrix column vectors and parallelogram.

Let's go over the key statements below.

- a** Creates a two-dimensional array representing matrix $\mathbf{A}$.
 - b** $\mathbf{A}[:, 0]$ extracts the first column from matrix $\mathbf{A}$ — that is, the matrix's left column. “:” means “all rows,” and 0 means “column 0” (Python indexing starts at 0). Similarly, $\mathbf{A}[:, 1]$ extracts the second column from matrix $\mathbf{A}$.
 - c** This prepares the four points needed to draw a parallelogram. Then, `np.array([P1, P2, P3, P4])` packs these four points into an array, forming a sequence of points ready to be plotted as a shape.
 - d** First, `arrow()` draws the first vector $\mathbf{a}_1$, starting from the origin. The first four parameters are the arrow's position: where it starts and where it points. Here, `*origin` unpacks the origin's coordinates (equivalent to writing 0, 0), and `*a1` unpacks the arrow's size along each direction. `head_width=0.2` makes the arrowhead a bit wider, `length_includes_head=True` means the arrow's length includes the arrowhead, and `color='red'` sets the arrow's color to red.
- Then, using the same method, draw the second vector, except this time the color is green.
- e** The `ax.fill()` function draws a closed region, such as the parallelogram in this example. What's passed in here are the coordinates of the four points: `parallelogram[:, 0]` represents the x-coordinates of all the points, and `parallelogram[:, 1]` represents the y-coordinates of all the points. `edgecolor='k'` sets the border color to black (k is short for black), `linestyle='--'` makes the border a dashed line, `alpha=0.3` sets the transparency to 30%, and `facecolor='yellow'` sets the fill color to yellow.

Code 1. Visualizing the Parallelogram Corresponding to a 2×2 Square Matrix |  LA_04_01_01.ipynb

```

## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 定义一个 2x2 的矩阵 A
a A = np.array([[2, 1],
                [1, 3]])

## 提取两列向量`
b a1 = A[:, 0] # 第一列向量
  a2 = A[:, 1] # 第二列向量

## 构建平行四边形
c origin = np.array([0, 0])
  # 起点为原点
  P1 = origin
  P2 = a1
  P3 = a1 + a2
  P4 = a2
  parallelogram = np.array([P1, P2, P3, P4])

## 可视化
fig, ax = plt.subplots()

# 绘制向量箭头
d ax.arrow(*origin, *a1, head_width=0.2,
           length_includes_head=True, color='red')
  ax.arrow(*origin, *a2, head_width=0.2,
           length_includes_head=True, color='green')

# 绘制平行四边形
e ax.fill(parallelogram[:, 0], parallelogram[:, 1],
          edgecolor = 'k', linestyle = '--',
          alpha=0.3, facecolor='yellow')

# 图像装饰
ax.set_aspect('equal')
ax.set_xlim(-1, max(P3[0], P2[0], P4[0]) + 1)
ax.set_ylim(-1, max(P3[1], P2[1], P4[1]) + 1)
ax.axhline(0, color='k', lw=1)
ax.axvline(0, color='k', lw=1)
ax.grid(True, color = '0.88')

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please compute the determinants of the following diagonal square matrices, and draw on grid paper the parallelograms corresponding to these matrices' column vectors.

►
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

▶ $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

Q2. Please compute the determinants of the following upper triangular matrices, and draw on grid paper the parallelograms corresponding to these matrices' column vectors.

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 2 \\ 0 & 3 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 8 \\ 0 & 0 \end{bmatrix}$

Q3. Please compute the determinants of the following lower triangular matrices, and draw on grid paper the parallelograms corresponding to these matrices' column vectors.

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 2 & 3 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 8 & 0 \end{bmatrix}$

Q4. Please swap the column vectors in the matrices from Q1, Q2, and Q3, and recompute the determinants.

Q5. Please explore the connection between determinants and the cross product of vectors.